

Functional Analysis of Omics Data

Protein Interaction Network Analysis and Visualization

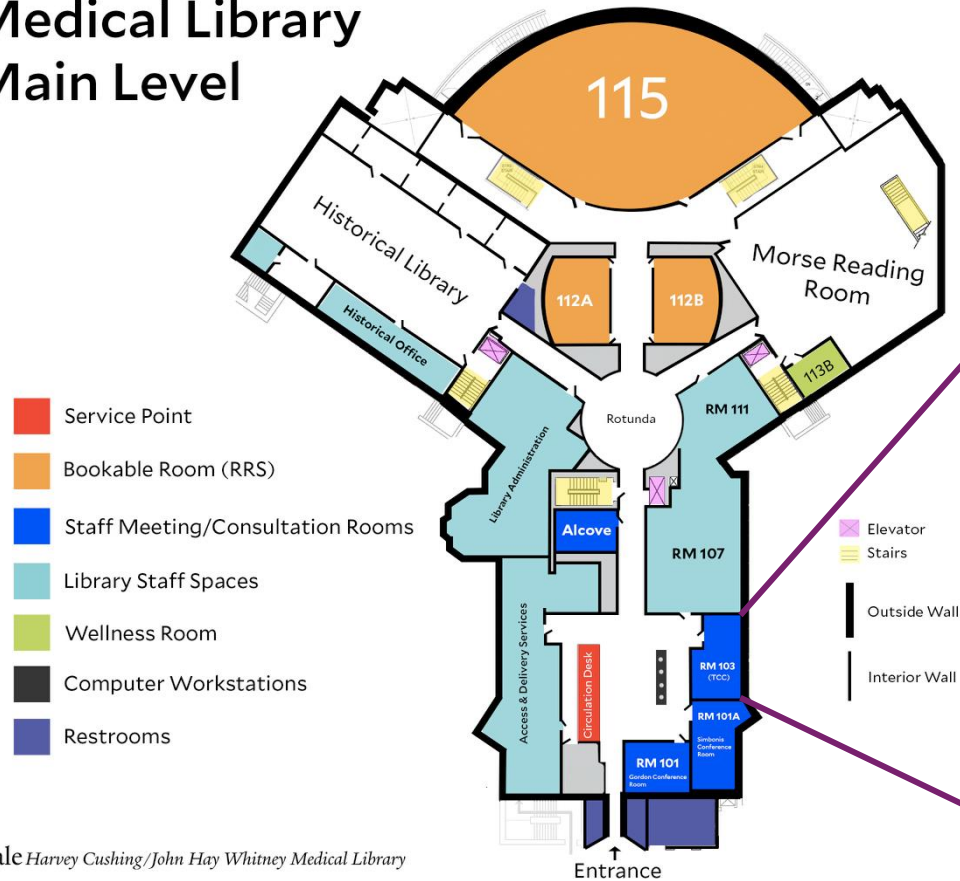
https://github.com/rologmilian/ppi_network

Rolando Garcia-Milian, MLS, AHIP
Bioinformatics Support Program
Rolando.milian@yale.edu



Bioinformatics Support Hub (BSH) supporting research since 2014

Medical Library Main Level



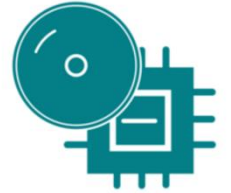
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Consultations & Collaborations



Training



Software & Hardware

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About

The Bioinformatics Support Program provides free access to state-of-the-art commercial bioinformatics software programs and databases for the mining, analysis, functional annotation, and visualization of Next Generation Sequencing data such as Partek Flow, QluCore Omics Explorer, Ingenuity Pathway Analysis, Ingenuity Variant Analysis MetaCore/MetaDrug, Clarivate Drug Intelligence, Biocyc, and TRANSFAC.

It also provides free individual and group instruction and consultation on how to use these software programs as well as many other public databases, data repositories and online tools such as those of the National Center for Biotechnology Information, European Bioinformatics Institute, The Cancer Genome Atlas, and others.

Who We Are



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Instruction and Training



Workshop series



Hands-on



Demos

September 10 – October 6, 2026

Bioinformatics Newsletter



SEP 8	Tue, 10:00am - 11:30am	In-Person	Medical School, SHM L 111, Cushing/Whitney Medical School	WAITLIST OPEN
Code with Clarity: AI-Powered Programming for Researchers In this training session, we will introduce Yale's Clarity AI system and other AI-powered coding tools designed to enhance your programming workflow. Through live demonstrations and examples, you'll... More				
■ Data				

SEP 8	Tue, 12:00pm - 12:30pm	Online	Medical School	20 SEATS LEFT
Finding Research Data at Yale Yale provides many rich data resources, so how do you find out if there is data somewhere that answers your research question? This interactive session introduces the biomedical and health sciences... More				
■ Data				

SEP 23	Wed, 10:00am - 12:00pm	In-Person	SHM L 111, Cushing/Whitney Medical School	6 SEATS LEFT
Mastering Differential Gene Expression Analysis with DESeq2 in Positron: From Demo to Your Data... Registration required Class 1 (you must attend Class 1 in order to take Class 2)* and In-Person Summary: This two-part workshop is designed to equip you with the essential skills for performing... More				
■ Bioinformatics				

SEP 24	Thu, 10:00am - 12:00pm	In-Person	SHM L 111, Cushing/Whitney Medical School	6 SEATS LEFT
Mastering Differential Gene Expression Analysis with DESeq2 in Positron: From Demo to Your Data... Registration required for Class 1 (you must attend Class 1 in order to take Class 2)* and In-Person Summary: This two-part workshop is designed to equip you with the essential skills for performing... More				
■ Bioinformatics				

SEP 10	Thu, 10:00am - 12:00pm	In-Person	Medical School, SHM L 111, Cushing/Whitney Medical School	11 SEATS LEFT
Network Visualization to Interpret Pathway Analysis Results with Enrichment Map Pathway (gene set) enrichment analyses often produce long lists of overlapping and redundant gene sets, which can make results difficult to interpret and act on. Enrichment Map addresses this by... More				
■ Bioinformatics				

SEP 16	Wed, 10:00am - 12:00pm	In-Person	Medical School, SHM L 111, Cushing/Whitney Medical School	14 SEATS LEFT
From Differential Expression to Mechanism: Protein-Protein Interaction Network Analysis... Differential expression analysis (e.g., DESeq2, limma) is powerful for identifying molecules whose expression changes across conditions. However, it often yields long gene lists that can be... More				
■ Bioinformatics				

SEP 29	Tue, 9:00am - 9:30am	In-Person	Medical School, Zoom Session -Medical (o...	14 SEATS LEFT
Webinar: QluCore Omics Explorer.Import, preprocess and analyze Olink® data Register here In this webinar presented by QluCore, you will learn how to go through the steps required to import and analyze Olink® data, from smaller to large protein panels. We will look at the... More				
■ Bioinformatics				

OCT 6	Tue, 10:00am - 11:00am	In-Person	Medical School, Zoom Session -Medical (o...	14 SEATS LEFT
Webinar: From Transcription Factors, Master Regulators and Disease Biomarkers to Omics Data... TRANSFAC® (Transcription Factor Database) is a comprehensive, manually curated database detailing eukaryotic transcription factors, their specific genomic binding sites, and DNA-binding profiles... More				
■ Bioinformatics				

SEP 17	Thu, 10:00am - 12:00pm	In-Person	Medical School, SHM L 111, Cushing/Whitney Medical School	14 SEATS LEFT
The Very Basics of the Unix Command Line This 2-hour session will introduce you to: - Navigating a *nix operating system with a terminal. - Basics of running a command. - Essential command line tools for reading and editing files. ... More				
■ Data				

SEP 22	Tue, 10:00am - 12:00pm	In-Person	Medical School, SHM L Cushing Center	14 SEATS LEFT
Bring Your Data: Hands-On Introduction to Data Preparation and Coding Have a dataset but not sure where to start? Or no dataset yet? Either way, we'd love to have you! Join us for a hands-on, beginner-friendly coding session led by library staff. In this interactive... More				
■ Data				

Yale Connect



Instruction and Training



Peer-to-peer teaching

Not professional teachers or instructors but member of the community

Cognitive congruence: awareness of current challenge of learners

MEDICAL REFERENCE SERVICES QUARTERLY
2022, VOL. 41, NO. 1, 13–25
<https://doi.org/10.1080/02763869.2022.2020568>



Peer Teaching as Bioinformatics Training Strategy: Incentives, Challenges, and Benefits

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Ann T. Chen^d, Razib Obaid^e , and Rolando Garcia-Milian^a 

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Main entrance of Yale School of Medicine



“Those having torches will pass them on to one another.”

Instruction and Training



Collaborations

10-sessions workshop series
30/ 67 preselected attendees



March 2026 Intro to NGS data analysis workshop **schedule** Every Tuesday from 9-12

Yale Connect



- Day 1. **Introduction to Bioinformatics Tools and Jupyter Notebooks** (Tue Mar 24)
- Day 2. **Project Management, Reproducibility, and Introduction to R with the Tidyverse** (Tue Mar 31)
- Day 3. **Experimental Design and Quality Control** (Tue Apr 7)
- Day 4. **Bulk RNA-seq Analysis (Single-Factor Design)** (Tue Apr 14)
- Day 5. **Bulk RNA-seq Analysis (Multi-Factor Design)** (Tue Apr 21)
- Day 6. **Pathway and Functional Enrichment Analysis** (Tue Apr 28)
- Day 7. **Single-Cell RNA-seq Analysis with Seurat (Single Sample)** (Tue May 5)
- Day 9. **Single-Cell RNA-seq Analysis with Seurat (Multiple Samples)** (Tue May 12)
- Day 10. **Somatic Variant Calling** (Tue May 26)

Instruction and Training



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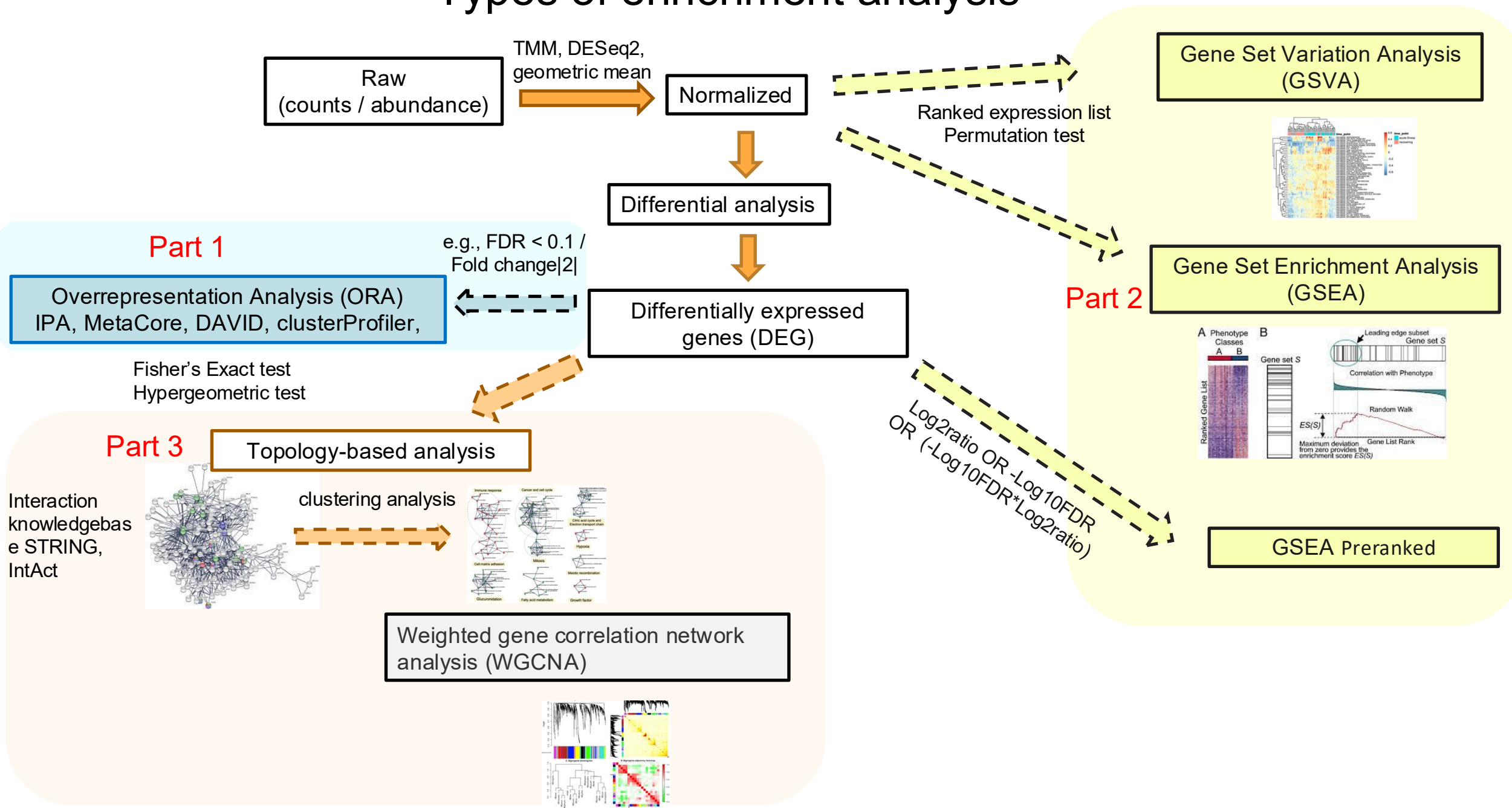


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Session Goals

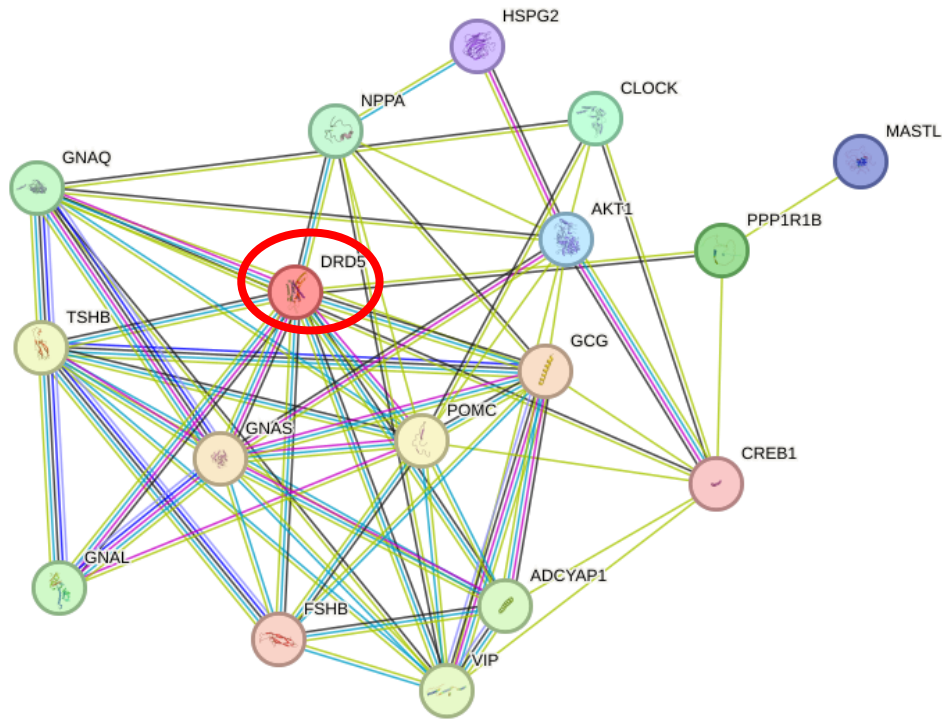
- Loading differential expression result to Cytoscape with the stringApp
- Protein-interaction (PPI) network enrichment analysis with stringApp, Enrichment Map, and Autoannotate
- PPI network clustering with stringApp and functional annotation
- Finding top hubs of network (cytoHubba app)
- Overlaying expression values on the network

Types of enrichment analysis



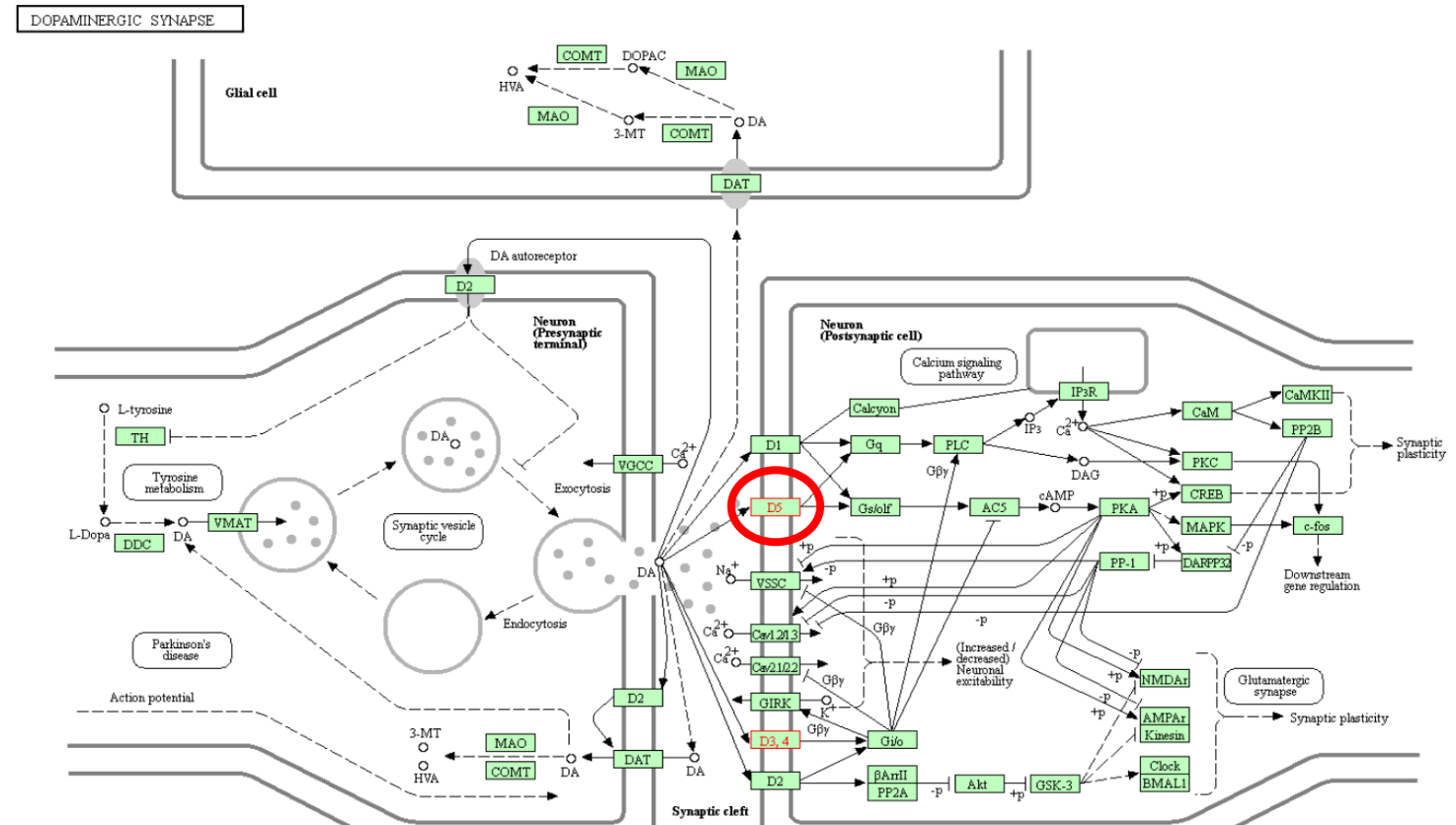
Pathway or network?

Dopamine receptor interaction network



STRING database

Dopaminergic synapse pathway



Kyoto Encyclopedia of Genes and Genomes (KEGG)

GO Annotations for human dopamine receptor

Total annotations: 66; showing: 1-66
Results count

Filter results

Total annotations: 66

User filters

– evidence_subset_closure_label: physical interaction evidence used in manual assertion

Your search is pinned to these filters

– document_category: annotation
– bioentity: UniProtKB:P21918

Ontology (aspect)

Organism

Type

Evidence

- ☒ (35) evidence used in automatic assertion
- ☒ (13) direct assay evidence used in manual assertion
- ☒ (13) experimental evidence
- ☒ (8) author statement supported by traceable reference used in manual assertion
- ☒ (6) biological aspect of ancestor evidence used in manual assertion
- ☒ (6) similarity evidence used in manual assertion
- ☒ (4) author statement without traceable support used in manual assertion

GO class (including "regulates")

GO class (excluding "regulates")

GO class (direct)

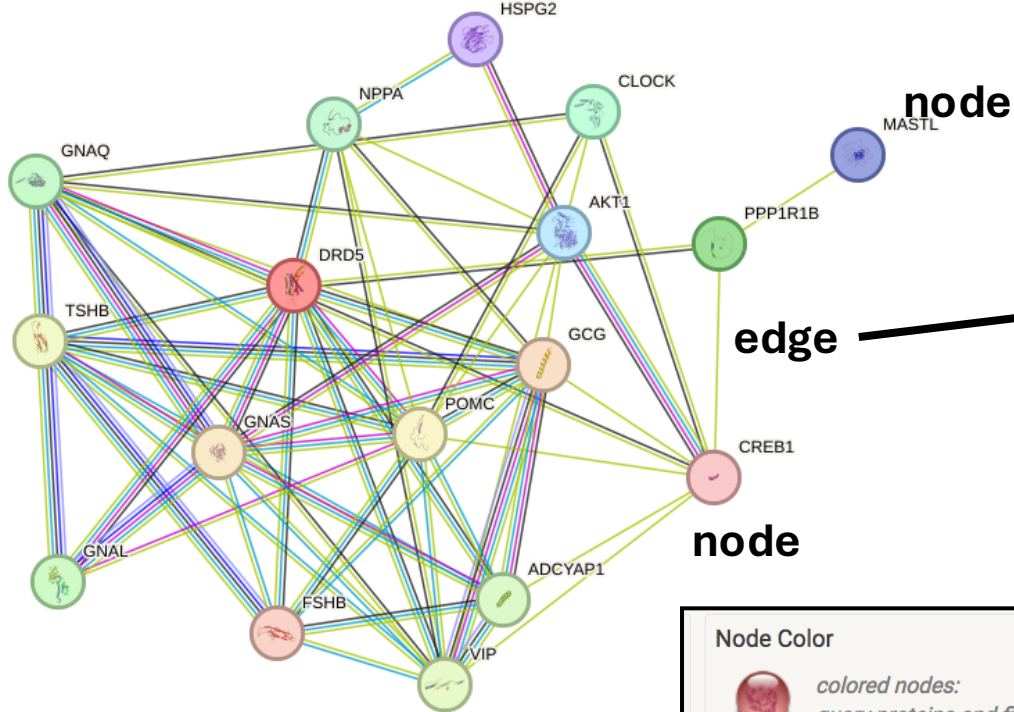
Annotation qualifier

«First <Prev Next> Last» Download (up to 100000)

☐ Gene/product	name	qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference
☐ DRD5	D(1B) dopamine receptor		positive regulation of MAPK cascade		GO_Central	Homo sapiens	IBA	PANTHER:PTN001205212 UniProtKB:P13945 more...	adrenergic receptor-related g-protein coupled receptor pthr24248	protein		GO_REF:0000033
☐ DRD5	D(1B) dopamine receptor		dopamine neurotransmitter receptor activity, coupled via Gs		GO_Central	Homo sapiens	IBA	PANTHER:PTN008610765 MGI:MGI:94927	adrenergic receptor-related g-protein coupled receptor pthr24248	protein		GO_REF:0000033
☐ DRD5	D(1B) dopamine receptor		G protein-coupled receptor activity		GO_Central	Homo sapiens	IBA	PANTHER:PTN002798190 MGI:MGI:106673 more...	adrenergic receptor-related g-protein coupled receptor pthr24248	protein		GO_REF:0000033
☐ DRD5	D(1B) dopamine receptor		plasma membrane		GO_Central	Homo sapiens	IBA	PANTHER:PTN002798190 RGD:2059 more...	adrenergic receptor-related g-protein coupled receptor pthr24248	protein		GO_REF:0000033
☐ DRD5	D(1B) dopamine receptor		adenylate cyclase-activating adrenergic receptor signaling pathway		GO_Central	Homo sapiens	IBA	PANTHER:PTN001205212 UniProtKB:P08588 more...	adrenergic receptor-related g-protein coupled receptor pthr24248	protein		GO_REF:0000033
☐ DRD5	D(1B) dopamine receptor		G protein-coupled dopamine receptor signaling pathway		GO_Central	Homo sapiens	IBA	PANTHER:PTN008610765 RGD:2523 UniProtKB:P21918	adrenergic receptor-related g-protein coupled receptor pthr24248	protein		GO_REF:0000033
☐ DRD5	D(1B) dopamine receptor		response to amphetamine		Ensembl	Homo sapiens	IEA	UniProtKB:Q8BLD9 ensembl:ENSMUSP00000039691	adrenergic receptor-related g-protein coupled receptor pthr24248	protein	UniProtKB:P21918	GO_REF:0000107
☐ DRD5	D(1B) dopamine receptor		regulation of systemic arterial blood pressure by vasopressin		Ensembl	Homo sapiens	IEA	UniProtKB:Q8BLD9 ensembl:ENSMUSP00000039691	adrenergic receptor-related g-protein coupled receptor pthr24248	protein	UniProtKB:P21918	GO_REF:0000107
☐ DRD5	D(1B) dopamine receptor		norepinephrine-epinephrine vasoconstriction involved in regulation of systemic arterial blood pressure		Ensembl	Homo sapiens	IEA	UniProtKB:Q8BLD9 ensembl:ENSMUSP00000039691	adrenergic receptor-related g-protein coupled receptor pthr24248	protein	UniProtKB:P21918	GO_REF:0000107
☐ DRD5	D(1B) dopamine receptor		adenylate cyclase-activating G protein-coupled receptor signaling pathway		Ensembl	Homo sapiens	IEA	UniProtKB:Q8BLD9 ensembl:ENSMUSP00000039691	adrenergic receptor-related g-protein coupled receptor pthr24248	protein	UniProtKB:P21918	GO_REF:0000107
☐ DRD5	D(1B) dopamine receptor		mating behavior		Ensembl	Homo sapiens	IEA	UniProtKB:Q8BLD9 ensembl:ENSMUSP00000039691	adrenergic receptor-related g-protein coupled receptor pthr24248	protein	UniProtKB:P21918	GO_REF:0000107

https://amigo.geneontology.org/amigo/gene_product/UniProtKB:P21918

STRING network structure



Nodes:

Network nodes represent proteins

splice isoforms or post-translational modifications are collapsed, i.e. each node represents all the proteins produced by a single, protein-coding gene locus.

Edges:

Edges represent protein-protein associations

associations are meant to be specific and meaningful, i.e. proteins jointly contribute to a shared function; this does not necessarily mean they are physically binding each other.

Node Color



*colored nodes:
query proteins and first shell of interactors*



*white nodes:
second shell of interactors*

Node Content



*empty nodes:
proteins of unknown 3D structure*



*filled nodes:
some 3D structure is known or predicted*

Known Interactions



from curated databases



experimentally determined

Predicted Interactions



gene neighborhood



gene fusions



gene co-occurrence

Others



textmining



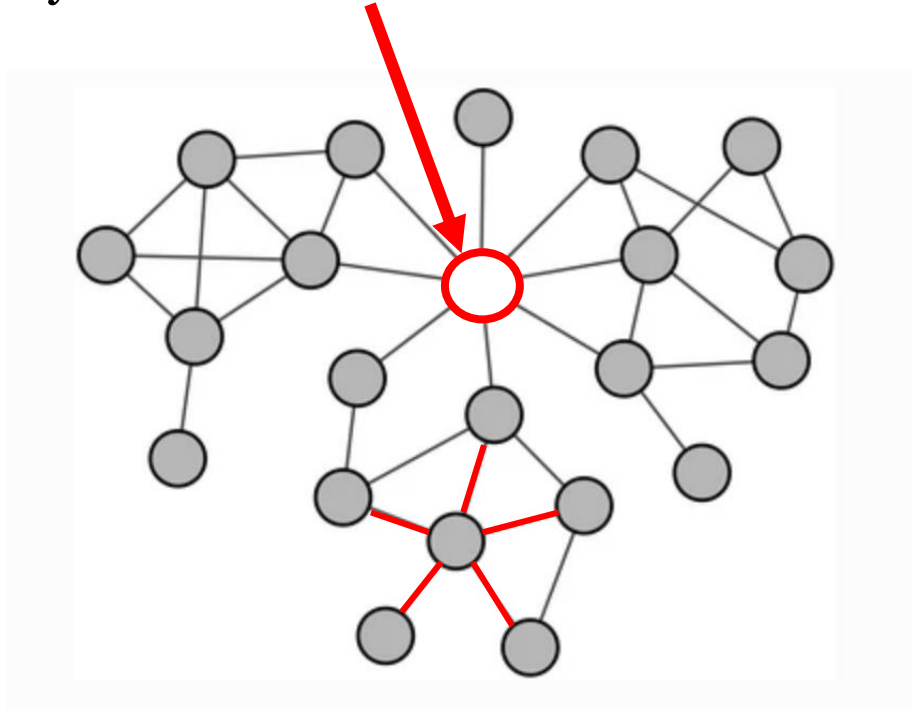
co-expression



protein homology

Hubs – Betweenness centrality

Removal of a few **key hubs** disconnects the system into small isolated node clusters:

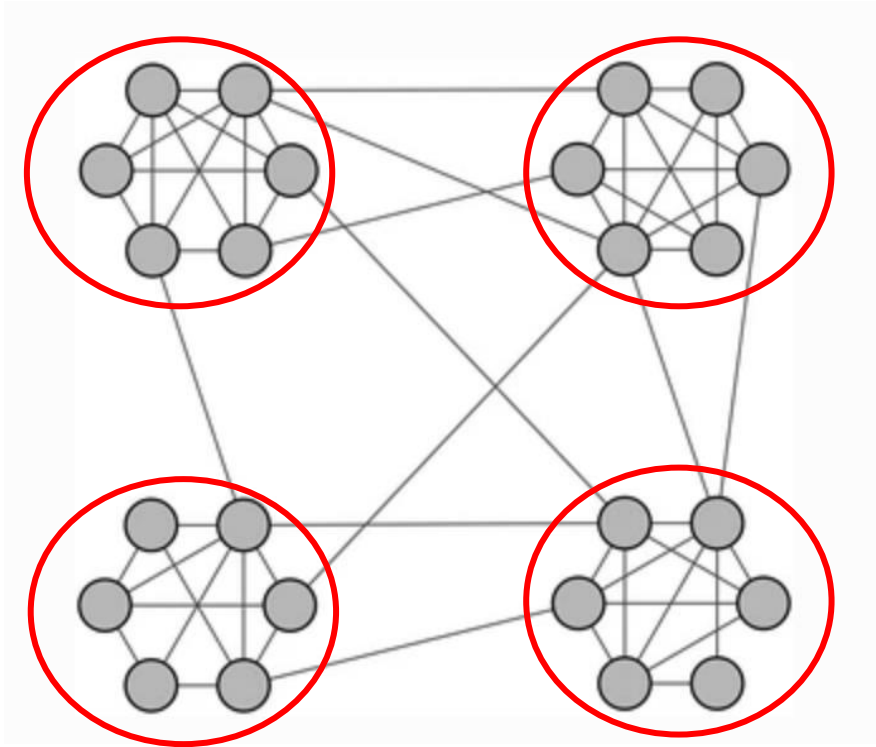


Betweenness centrality: role of a node in allowing information to pass from one part of the network to the other – shorter path

Degree: number of connections of a node (5)

Clusters

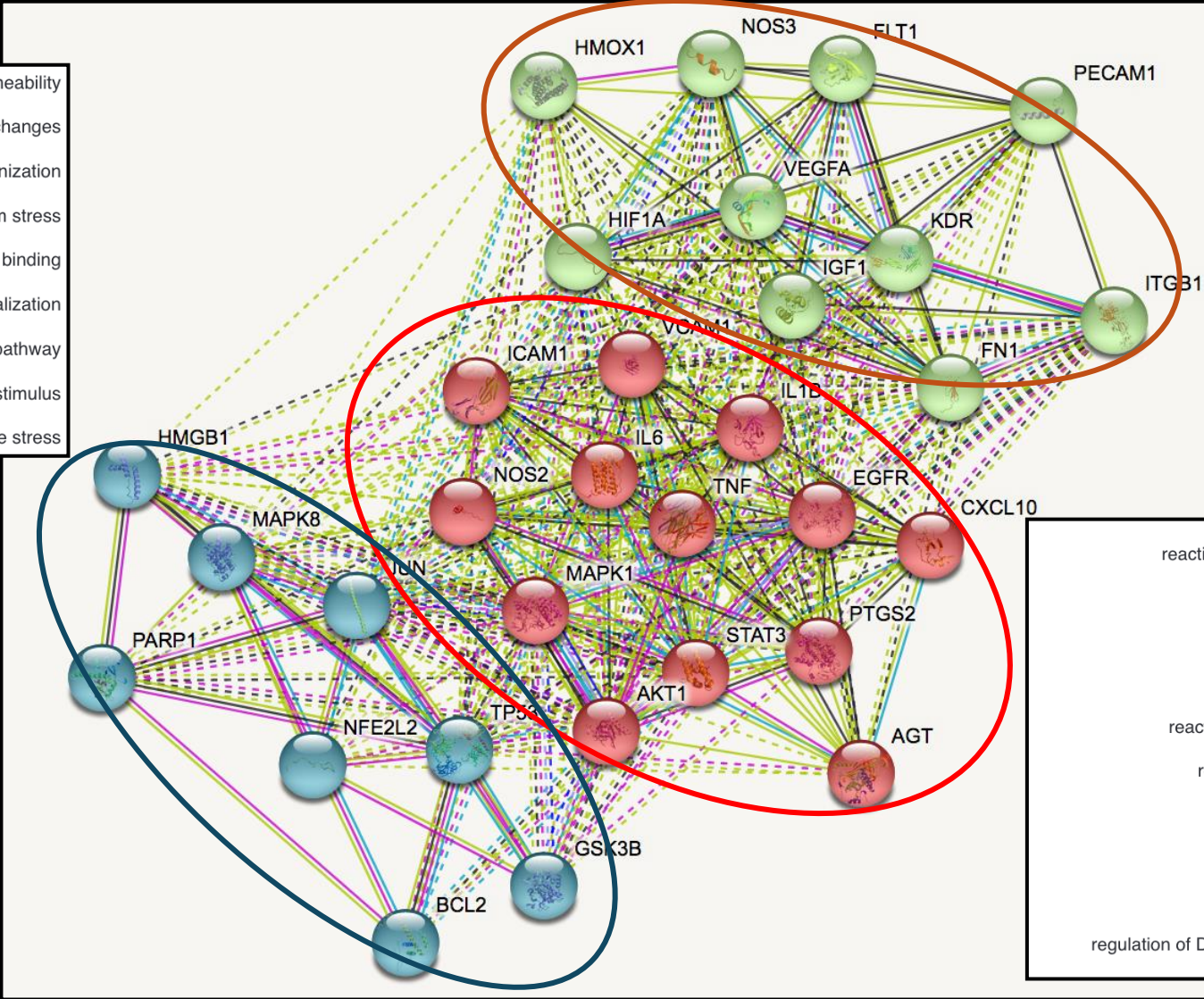
A subset in a graph with common characteristics or related to each other in particular manner



Modularity: A network with four distinct modules, each of which is highly connected but only lightly connected to other modules

STRING MCL Clustering

- regulation of membrane permeability
- apoptotic mitochondrial changes
- mitochondrial membrane organization
- response to endoplasmic reticulum stress
- regulation of binding
- positive regulation of cellular protein localization
- regulation of apoptotic signaling pathway
- cellular response to external stimulus
- response to oxidative stress



- vascular endothelial growth factor receptor signaling pathway
- platelet degranulation
- tissue migration
- regulation of vasculature development
- ameboidal-type cell migration
- angiogenesis
- positive regulation of cell motility
- epithelial cell proliferation
- cell-substrate adhesion
- leukocyte migration

- reactive nitrogen species metabolic process
- acute inflammatory response
- muscle cell proliferation
- cellular response to biotic stimulus
- reactive oxygen species metabolic process
- response to molecule of bacterial origin
- gliogenesis
- cellular response to drug
- positive regulation of cell motility
- regulation of DNA-binding transcription factor activity

The Markov Cluster Algorithm, aka the MCL algorithm

Inflation parameter effect

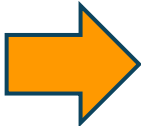


Inflation Value	Granularity	Cluster Count	Cluster Size
Low (1.2–1.4)	Coarse	Fewer	Larger
Default (~2.0)	Balanced	Moderate	Medium
High (4.0–6.0+)	Fine	More	Smaller

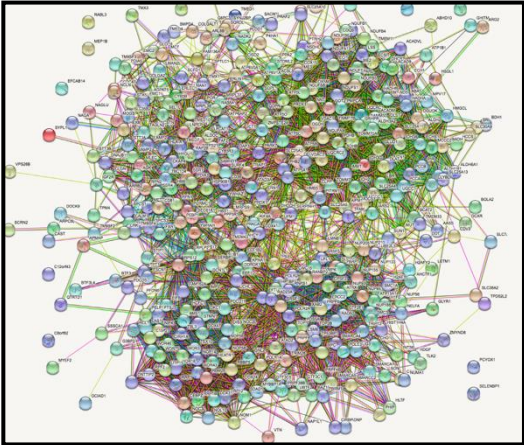
Interaction network analysis

User list

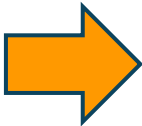
A
Gene symbol
HBB
VTN
ATP5L
HBA1
PBRM1
MCCC1
TIMM9
TMED4
GTF3C1
EIF4EBP1
STMN1
HYPK
TIMM13
SF3B5
VDAC3
MCCC2
PFDN4



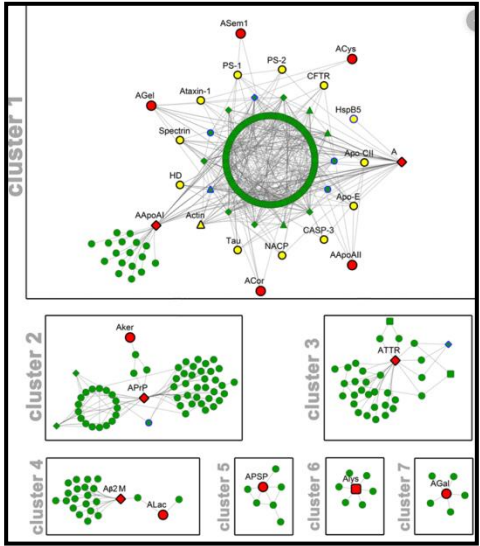
PPI Database



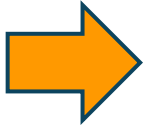
STRING
BioGRID
IntAct



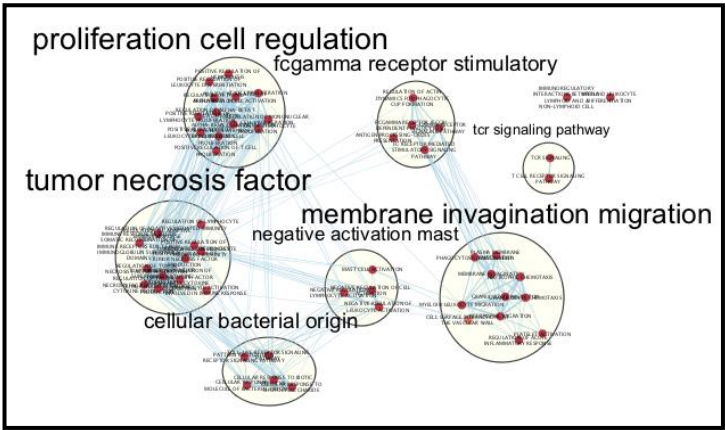
Cluster analysis



STRING-MCL
MCODE
ClusterONE
clusterMaker



Cluster annotation



EnrichmentMap

DEMO

GSE125583

RNA from fusiform gyrus of Alzheimer's or neurologically normal post-mortem tissue.

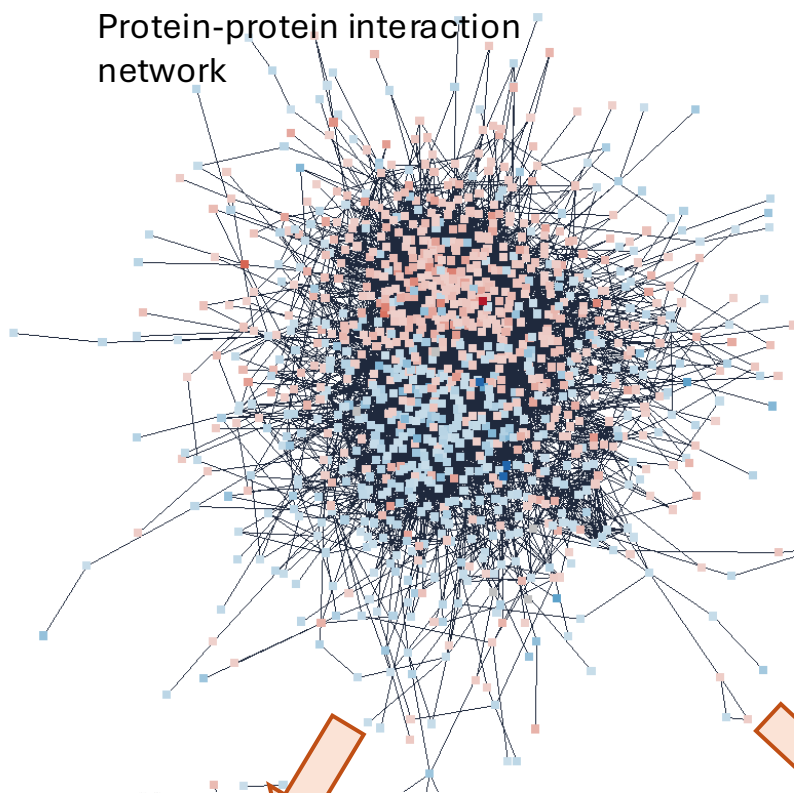
DESeq2

- female AD vs female control
- Male AD vs male control  Abs Log2 foldchange > 0.5, padj<0.05
(2163 DEGs)

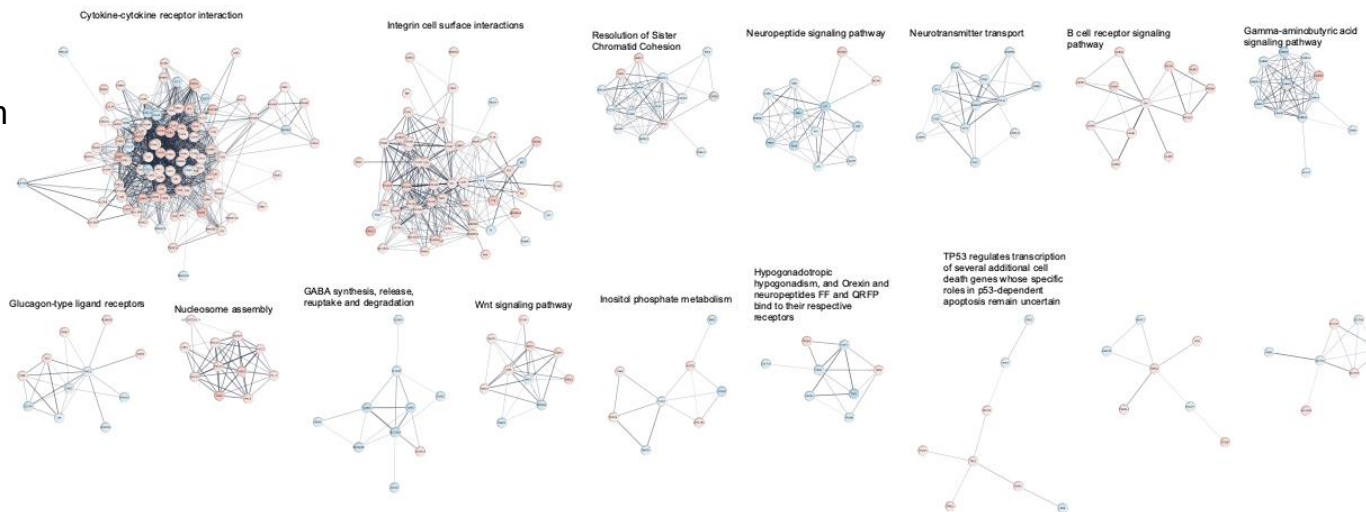
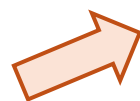
Cytoscape stringApp

DEMO

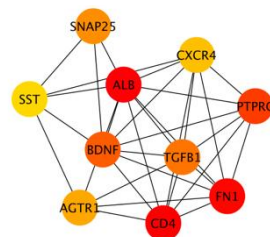
Protein-protein interaction network



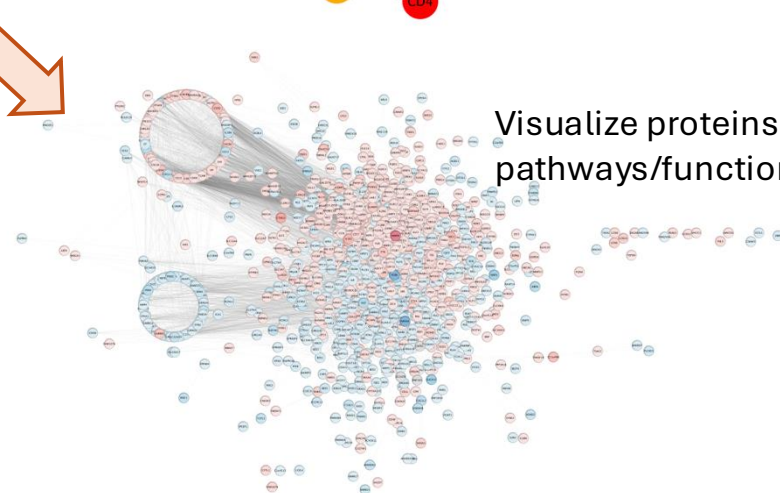
Clustering functional annotation



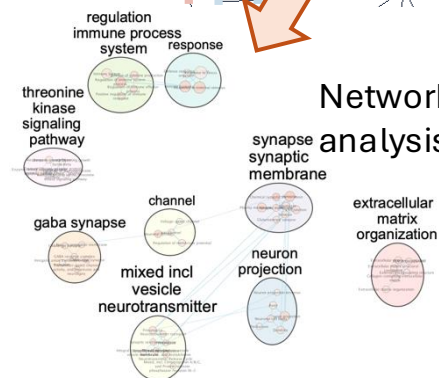
Identify hubs



Visualize proteins in pathways/functions



Network enrichment analysis



STRING database Gene Set Enrichment Analysis

Protein by name >

Multiple proteins >

Proteins by sequences >

Proteins with Values/Ranks >

Protein families ("COGs") >

Pathway / Process / Disease >

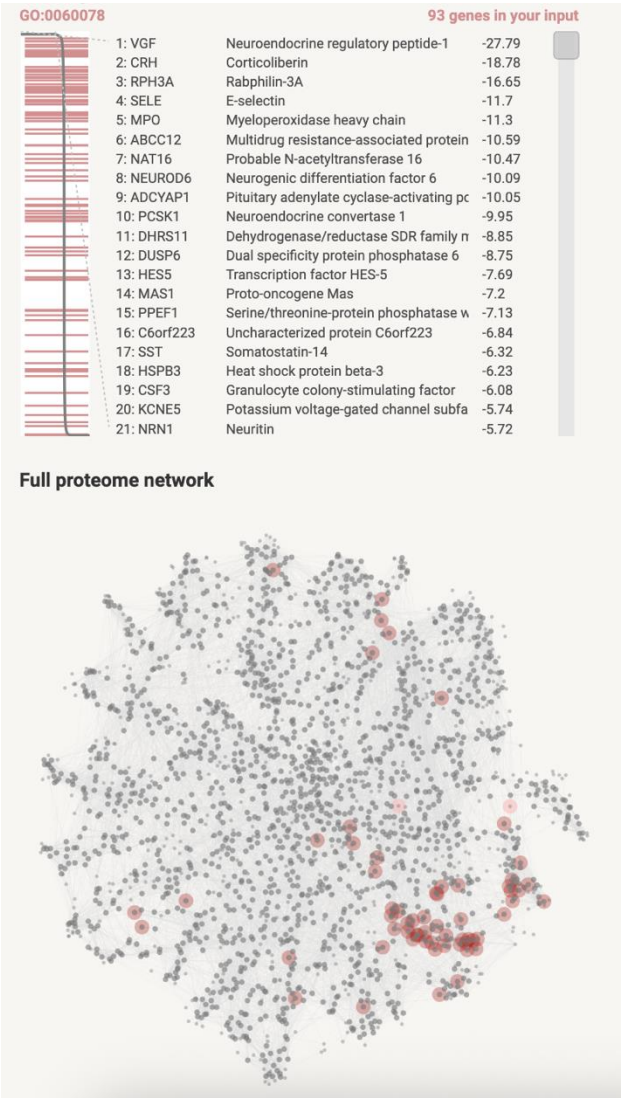
Add organism >

Organisms >

STRING chat New >

Examples >

Random entry >



explain columns

Biological Process (Gene Ontology)					
GO-term	description	* enrichment score	direction	count in set/pathway	false discovery rate
GO:0060078	Regulation of postsynaptic membrane potential	0.00546757	top of input	93 of 107	0.0067 (ks)
GO:0032984	Protein-containing complex disassembly	0.00698905	top of input	127 of 137	0.0040 (ks)
GO:0140053	Mitochondrial gene expression	0.0232891	top of input	141 of 143	0.0034 (ks)
GO:0051650	Establishment of vesicle localization	0.0246389	top of input	165 of 169	0.0043 (ks)
GO:0051648	Vesicle localization	0.0276273	top of input	176 of 180	0.00011 (ks)
GO:0070972	Protein localization to endoplasmic reticulum	0.0280691	top of input	66 of 66	0.0034 (ks)
GO:0002181	Cytoplasmic translation	0.0285379	bottom of input	122 of 123	6.44e-05 (ks)
GO:0033108	Mitochondrial respiratory chain complex assembly	0.0336414	top of input	97 of 98	0.00027 (ks)
GO:0032543	Mitochondrial translation	0.0413337	top of input	110 of 112	2.27e-05 (ks)
GO:0008306	Associative learning	0.0448071	top of input	79 of 83	0.0027 (ks)
GO:0009260	Ribonucleotide biosynthetic process	0.0450822	top of input	200 of 211	0.0010 (ks)
GO:0009260	Ribonucleotide biosynthetic process	0.0450822	top of input	200 of 211	0.0010 (ks)

<https://version-12-0.string-db.org/cgi/globalenrichment?networkId=bZI5sov0bZ4N>

Examples of data visualizations



National Library of Medicine
National Center for Biotechnology Information

 **Search**



PubMed Central®

network[capt] AND string[capt] AND cluster[capt]

Search in PMC

[Journal List](#) | [User Guide](#)

Additional references and tutorials

- Dančík, V., Basu, A., Clemons, P. (2013). Properties of Biological Networks. In: Prokop, A., Csukás, B. (eds) Systems Biology. Springer, Dordrecht. https://doi.org/10.1007/978-94-007-6803-1_5
- Chin CH, Chen SH, Wu HH, Ho CW, Ko MT, Lin CY. cytoHubba: identifying hub objects and sub-networks from complex interactome. BMC Syst Biol. 2014;8 Suppl 4(Suppl 4):S11. doi: 10.1186/1752-0509-8-S4-S11. Epub 2014 Dec 8. PMID: 25521941; PMCID: PMC4290687
- Isserlin R, Merico D, Voisin V, Bader GD. Enrichment Map - a Cytoscape app to visualize and explore OMICs pathway enrichment results. F1000Res. 2014 Jul 1;3:141. doi: 10.12688/f1000research.4536.1. PMID: 25075306; PMCID: PMC4103489.
- Kucera M, Isserlin R, Arkhangorodsky A, Bader GD. AutoAnnotate: A Cytoscape app for summarizing networks with semantic annotations. F1000Res. 2016 Jul 15;5:1717. doi: 10.12688/f1000research.9090.1. PMID: 27830058; PMCID: PMC5082607
- Shannon P, Markiel A, Ozier O, Baliga NS, Wang JT, Ramage D, Amin N, Schwikowski B, Ideker T. Cytoscape: a software environment for integrated models of biomolecular interaction networks. Genome Res. 2003 Nov;13(11):2498-504. doi: 10.1101/gr.1239303. PMID: 14597658; PMCID: PMC403769
- Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryary F, Hachilif R, Gable AL, Fang T, Doncheva NT, Pyysalo S, Bork P, Jensen LJ, von Mering C. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. Nucleic Acids Res. 2023 Jan 6;51(D1):D638-D646. doi: 10.1093/nar/gkac1000. PMID: 36370105; PMCID: PMC9825434

Please help us improve your experience



Thank you!

Rolando Garcia-Milian, MLS, AHIP

Bioinformatics Support Program

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